



Smart Agriculture Analysis

*An Exploratory Data Analysis of Crop Suitability
Across Karnataka's Agro-Climatic Regions*

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Dataset: karnataka_crop_dataset_V2.csv | Tools: Python · Pandas · Seaborn · Matplotlib

1. Executive Summary

Karnataka's agricultural landscape spans four distinct agro-climatic zones mainly Coastal, Malnad, North Karnataka, and South Karnataka each defined by unique soil profiles, rainfall patterns, and crop suitability. This report presents a structured exploratory data analysis (EDA) of 21,960 agronomic records drawn from a Karnataka-specific crop recommendation dataset, covering 39 crop types, 31 districts, and 6 soil classifications.

The analysis was conducted in Python using Pandas, Seaborn, and Matplotlib, and is organised around five core questions:

1. How crop recommendations distribute across seasons and regions?
2. How rainfall varies by zone?
3. What the top crops require in terms of soil conditions?
4. What macro-nutrients (Nitrogen, Phosphorous, and Potassium) differ by soil type?
5. Which crops impose the highest and lowest nutrient demand on the soil?

Key Headline Findings

- Kharif (monsoon) crops dominate recommendations in all four regions, particularly in Coastal Karnataka where rainfall is highest.
- The dataset has no missing values across any of its 12 features, confirming its quality for analysis.
- Phosphorous content remains remarkably stable (~45 kg/ha) regardless of soil type which is an agronomically significant pattern.
- Sugarcane demands the highest combined N-P-K from the soil; Moth Beans (Madke Kalu) the least.
- Among kitchen staples, Coconut is the largest soil-nutrient consumer; Ragi the smallest.
- Most crops can be grown under low rainfall conditions, suggesting widespread dependence on irrigation infrastructure in Karnataka.

2. Introduction

Agriculture remains Karnataka's economic backbone, engaging over 56% of the state's workforce and accounting for a significant share of its GDP. However, the state's agricultural productivity is constrained by high variability in soil quality, rainfall distribution, and crop suitability across its agro-climatic diversity. Data-driven decision-making using soil and climatic profiles to match the right crop to the right land offers a scalable path to improved yield and resource efficiency.

This project applies exploratory data analysis to a curated crop recommendation dataset for Karnataka, translating raw agronomic records into interpretable patterns. The analysis is intended to serve agronomists, extension officers, and data science practitioners as a reference for understanding the agro-climatic drivers behind crop recommendation density across the state.

Research Questions

- **Q1:** How are crop recommendations distributed across seasons and agro-climatic regions?
- **Q2:** What are the average rainfall requirements by region and farming season?
- **Q3:** What are the top recommended crops, and what are their soil pH and rainfall profiles?
- **Q4:** How do macro-nutrient (N-P-K) profiles differ across soil types?
- **Q5:** Which crops have the highest and lowest total nutrient demand?

3. Dataset & Methodology

3.1 Dataset Overview

The dataset used is ‘karnataka_crop_dataset_V2.csv’, a structured agronomic recommendation dataset specific to Karnataka, India, obtained from Kaggle under the title: "Smart Agriculture Analytics: Crop Requirement & Soil Analysis", which was a BE pre-final year project of Megha R Ganesh from DSCE Bangalore.

Attribute	Value
Total Records	21,960
Unique Crops	39
Districts Covered	31
Agro-Climatic Regions	4 (Coastal, Malnad, North Karnataka, South Karnataka)
Soil Types	6 (Loamy, Red, Black, Sandy, Laterite, Alluvial)
Farming Seasons	3 (Kharif, Rabi, Zaid)
Missing Values	None (0 across all 12 columns)

3.2 Feature Reference

Feature	Type	Description
N	Continuous	Nitrogen content in soil (kg/ha)
P	Continuous	Phosphorous content in soil (kg/ha)
K	Continuous	Potassium content in soil (kg/ha)
temperature	Continuous	Average ambient temperature (°C)
humidity	Continuous	Relative humidity (%)
ph	Continuous	Soil pH (acidity/alkalinity scale)
rainfall	Continuous	Average seasonal rainfall (mm)
soil_type	Categorical	Soil classification (Loamy, Red, Black, Sandy, Laterite, Alluvial)
season	Categorical	Farming season (Kharif, Rabi, Zaid)
district	Categorical	District in Karnataka
region	Categorical	Agro-climatic zone
label	Categorical	Recommended crop-target variable

3.3 Data Cleaning Steps

The dataset was verified to be clean prior to EDA, requiring only minor pre-processing:

- All categorical columns (soil type, season, district, region, label) were stripped of leading and trailing whitespace using `str.strip()`.
- A full null-value audit confirmed zero missing entries across all 12 columns.
- Data types were validated: 7 continuous numerical features and 5 string-categorical features.

3.4 Summary Statistics for Numerical Features

Stat	N	P	K	Temp (°C)	Humidity (%)	pH	Rainfall (mm)
Mean	87.4	44.2	81.8	26.7	64.9	6.44	475.7
Std Dev	52.0	13.4	46.0	3.8	13.0	0.51	629.4
Min	15.0	20.0	15.0	15.0	30.0	4.53	53.7
Max	279.8	100.0	200.0	38.0	95.0	8.00	3678.9

⚡ *The high standard deviation in rainfall (629 mm vs mean 476 mm) signals extreme right-skew driven by Coastal Karnataka’s monsoon-heavy zones.*

4. Exploratory Data Analysis: Findings

4.1 Regional & Seasonal Crop Distribution

A grouped bar chart of crop recommendation counts by agro-climatic region, and farming season revealed a clear seasonal hierarchy consistent across all four zones.

- Kharif (monsoon season, June–October) dominates crop recommendations across every region.
- Rabi (post-monsoon / winter, November–February) is the second-most recommended season.
- Zaid (summer, March–May) has the fewest crop recommendations, constrained by heat and water availability.

⚡ *The dominance of Kharif crops is not merely a cultural preference but reflects Karnataka’s dependence on the Southwest Monsoon, which delivers approximately 75% of the state’s annual rainfall.*

A secondary plot of the top 5 most recommended crops overall (irrespective of season) shows groundnut, banana, maize, mango, and papaya as frequently appearing crop recommendations in the dataset which is a mix of commercial, horticultural, and food security crops.

4.2 Rainfall Analysis by Region and Season

Average seasonal rainfall was computed by region and season and plotted as a grouped bar chart. The findings confirm well-known agro-climatic gradients across Karnataka:

- Coastal Karnataka records the highest average rainfall, particularly during Kharif season, driven by the Arabian Sea branch of the Southwest Monsoon.
- Malnad (the Western Ghats foothills) follows closely, benefiting from orographic rainfall.

- North Karnataka and South Karnataka show considerably lower average rainfall, with North Karnataka exhibiting the greatest dependency on Rabi crops as a result.

⚡ *The rainfall outlier range (up to 3,679 mm) is almost certainly attributable to Coastal Karnataka records and aligns with known annual rainfall data for coastal districts like Dakshina Kannada and Udupi.*

A rainfall distribution boxplot for the top 10 crops showed that most crops cluster in the 100–500 mm range, with high-rainfall outliers present in almost all categories. The exception is mango, which appears to tolerate the widest rainfall range of any top crop, making it suitable across multiple agro-climatic zones.

4.3 Crop-Level Soil pH & Rainfall Requirements

Soil pH and rainfall were analysed for the top 5 most recommended crops using both a combined scatter plot and a per-crop FacetGrid of individual scatter panels, with a reference line drawn at pH 7.0 (chemically neutral).

- Groundnut shows the tightest clustering around a specific pH band (mildly acidic, 5.5 - 6.5), making it the most pH-sensitive of the top 5 crops.
- Mango is the least pH-sensitive, with data points scattered across the full pH range (5.5 – 7.5).
- Banana consistently plots below pH 7.0, confirming its preference for slightly acidic soil conditions.
- The base of each crop's scatter cloud sits near zero rainfall (50–150 mm), with high-rainfall points appearing as outliers in all crops except mango. This pattern strongly suggests that most crop cultivation in the dataset relies on irrigation rather than rain-fed agriculture.

⚡ *The irrigation dependency signal is agronomically significant: it means that 'rainfall' in this dataset may function as an ecological threshold variable rather than a direct cultivation input for most crops.*

4.4 Macro-Nutrient (N-P-K) Profiles by Soil Type

Average Nitrogen (N), Phosphorous (P), and Potassium (K) content was computed per soil type and plotted as a grouped bar chart.

- Black soil has the lowest macro-nutrient holding capacity across all three nutrients counterintuitive given its reputation for water retention, but consistent with its low organic matter content in certain Karnataka regions.
- Laterite and Alluvial soils hold the highest average N-P-K levels, reflecting their richer mineralogical profiles.
- Sandy and Red soils occupy the mid-range for N and K.

⚡ *The most striking finding: Phosphorous (P) remains consistently near 45 kg/ha regardless of soil type. This near-constant P across 6 soil classifications is agronomically unusual and warrants further investigation, it may reflect standardised fertiliser recommendations in the source data rather than naturally occurring soil P levels.*

4.5 Heatmap Analysis: Districts and Soil Types vs Crops

Two heatmaps were generated: one mapping all 31 Karnataka districts against all 39 crop types, and a second mapping 6 soil types against crops.

District-Crop Heatmap: Certain districts show strong crop specialisation (concentrated green bands in specific crop columns), while others show broader multi-crop suitability. This reflects the influence of local agro-climatic conditions, irrigation infrastructure, and historical cropping patterns.

Soil-Crop Heatmap: Red, Loamy, and Black soils collectively account for the highest crop recommendation density. This is consistent with Karnataka's land composition: Red lateritic soils cover approximately 60% of the state's cultivable area.

⚡ *The sparse Alluvial and Sandy columns in the soil heatmap suggest this soil types appear in fewer districts, or that fewer crop types are recommended for them and **not necessarily** that they are **less productive**.*

4.6 Feature Correlation Matrix

A Pearson correlation heatmap was generated for all 7 numerical features (N, P, K, temperature, humidity, pH, rainfall). The key observations were as follows:

- N and K show moderate positive correlation crops that require high nitrogen tend to also require high potassium, consistent with known plant physiology.
- P shows weak correlation with all other features, supporting the plateau finding in Section 4.4.
- Temperature and humidity show mild negative correlation, as expected (hotter periods tend to be drier in Karnataka).
- Rainfall shows low correlation with most soil nutrient features, reinforcing that rainfall in this dataset is better understood as an ecological classifier than a direct agronomic input.

4.7 Nutrient Demand Rankings Across All 39 Crops

Total macro-nutrient demand (N+P+K) was computed for all 39 crops and plotted as a stacked horizontal bar chart, sorted ascending.

- Sugarcane tops the nutrient demand chart with the highest combined N-P-K absorption of all crops in the dataset consistent with its status as one of the most soil-depleting commercial crops globally.
- Moth Beans (Madke Kalu, ಮಡಕೆ ಕಾಳು) record the lowest combined nutrient demand, making them an ideal rotation or intercrop for soil recovery.
- Legumes as a category (blackgram, horsegram, chickpea, groundnut) cluster in the lower third of the nutrient demand chart reflecting their ability to fix atmospheric nitrogen, reducing fertiliser dependency.

⚡ *From a soil health management perspective, alternating sugarcane with moth beans or other legumes could significantly reduce net nutrient depletion over multiple crop cycles.*

4.8 Nutrient Demand: Kitchen Staples & Cereals

A focused analysis was run on 12 commonly consumed crops: onion, tomato, chilli, coconut, coffee, turmeric (kitchen staples) and rice, maize, wheat, ragi, jowar, groundnut (cereals and grains).

- Coconut is the largest nutrient consumer among this subset, driven primarily by high Potassium demand (consistent with its perennial fruiting cycle).
- Coffee and turmeric follow as mid-to-high nutrient crops.
- Ragi (finger millet) has the lowest N-P-K demand in this subset, reinforcing its status as a drought-resilient, low-input cereal well suited to Karnataka's rain-shadow zones.

⚡ *Ragi's combination of low nutrient demand, drought tolerance, and nutritional richness (high calcium and iron) makes it an analytically underappreciated crop relative to its agronomic value.*

5. Crop Suitability Reference Card

The table below summarises average soil and climatic requirements for the top 10 most recommended crops in the dataset, together with their most recommended soil type. This serves as a quick-reference agronomic cheat sheet for extension workers, educators, or digital agriculture applications.

Crop	Avg N	Avg P	Avg K	Rainfall (mm)	pH	Temp (°C)	Ideal Soil
Arecanut	139.2	54.1	155.2	742.1	6.1	26.6	Laterite
Banana	149.9	51.0	175.6	285.1	6.4	27.7	Red
Coconut	150.1	56.7	152.4	588.8	6.5	26.3	Laterite
Groundnut	19.6	49.8	70.0	388.6	6.5	29.0	Red
Guava	52.1	29.8	53.8	304.5	6.5	25.8	Red
Jowar	54.3	29.9	33.2	353.5	6.5	28.4	Black
Maize	95.9	46.0	50.3	476.1	6.5	24.8	Loamy
Mango	85.2	30.2	85.2	243.5	6.5	27.7	Red
Papaya	151.5	46.6	150.8	276.2	6.5	28.0	Red
Sunflower	71.1	47.2	52.3	348.8	6.6	26.1	Red

N, P, K values are in kg/ha. Rainfall is average seasonal rainfall in mm. Temperature is average ambient temperature in °C.

6. Scope & Limitations

This analysis is descriptive and exploratory. The following limitations should be kept in mind when interpreting the findings:

- Recommendation density ≠ productivity: The dataset counts how often a crop is recommended per agro-climatic profile, not how much it yields. A crop appearing 800 times is not necessarily 800 times more productive than one appearing 100 times.
- No temporal dimension: The dataset has no year or growing-cycle column. Year-over-year trends, climate change effects on crop suitability, or seasonal variation across years cannot be assessed.
- No socio-economic variables: Farmer income, market price, input costs, and labour availability are absent. These are critical real-world determinants of which crop a farmer grows.
- Irrigation ambiguity: The rainfall feature appears to represent ecological or potential rainfall rather than actual water received by crops. The irrigation dependency pattern observed in the scatter plots cannot be fully disentangled from the data as structured.
- Phosphorous plateau anomaly: The near-constant P across soil types (Section 4.4) warrants scrutiny. If this reflects standardised fertiliser prescription rather than measured soil P, it could indicate that the dataset was generated from a recommendation model rather than field surveys.

7. Recommendations & Future Work

The following extensions would significantly deepen the value of this analysis:

- **Predictive Modelling:** Train a multi-class crop recommendation classifier (Random Forest, XGBoost, or a simple neural network) using the 7 numerical features as inputs and the label column as output. This dataset is structured for ML out of the box.
- **Streamlit / Dash Dashboard:** Convert the Crop Suitability Card into an interactive web tool where users input their district, soil type, and season, and receive recommended crops ranked by suitability.
- **Yield Integration:** Merge with Karnataka's Agriculture Department's published yield data (available via data.gov.in) to move from recommendation density to actual productivity analysis.
- **Geodata Overlay:** Integrate IMD (India Meteorological Department) district-level rainfall geodata or ISRO Bhuvan satellite imagery to validate dataset rainfall distributions against observed actuals.
- **Cluster Analysis:** Apply K-Means or hierarchical clustering to identify natural groupings of crops by their combined soil-climate profile useful for identifying substitutable crops within the same agro-climatic niche.
- **Taluk-Level Granularity:** The current district-level resolution (31 districts) obscures intra-district variation. A taluk-level dataset would yield more actionable recommendations for farmers and extension officers.

Appendix: Visualisation Index

The following plots were generated during the analysis and are available in the Jupyter Notebook (Smart_Agri_Analysis.ipynb):

#	Visualisation Title	Chart Type
1	Distribution of Seasonal Crops Across Agro-Climatic Regions	Grouped Bar Chart
2	Top 5 Most Recommended Crops (Overall)	Horizontal Bar Chart
3	Average Seasonal Rainfall Across Regions	Grouped Bar Chart
4	Rainfall vs Soil pH for Top 5 Crops	Scatter Plot
5	Rainfall vs Soil pH per Crop (Individual Panels from Scatter Plot)	FacetGrid / Small Multiples
6	Rainfall Distribution Profiles — Top 10 Crops	Horizontal Boxplot
7	Average N-P-K Profile by Soil Type	Grouped Bar Chart
8	Heatmap: Crop Recommendations by District	Heatmap
9	Heatmap: Crop Recommendations by Soil Type	Heatmap
10	Feature Correlation Matrix	Correlation Heatmap
11	N-P-K Soil Requirements — All 39 Crops	Stacked Horizontal Bar
12	N-P-K Requirements — Kitchen Staples & Cereals	Stacked Horizontal Bar

All 39 crops in the dataset: arecanut, banana, barley, betel, blackgram, brinjal, capsicum, cardamom, cashew, chickpea, chilli, coconut, coffee, cotton, cowpea, ginger, groundnut, guava, horsegram, jowar, jute, lentil, maize, mango, marigold, mothbeans, onion, orange, papaya, peas, pepper, pomegranate, potato, rice, ragi, sapota, sugarcane, sunflower, tomato, turmeric, wheat.